

Digital Economy Intelligence Lab: A Comparative Diagnosis of Mexico's Digital Readiness in the Agricultural Sector

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Abstract

This report presents a comparative diagnosis of how prepared five economies are to participate in, generate, and capture value from the digital economy. The economies were assigned to Team 3 (Agriculture): Mexico, the Netherlands, Kenya, Argentina, and New Zealand. Using eight indicators from official sources (World Bank, ITU, UNCTAD, and WIPO) for the year 2023, a Digital Readiness Score (DRS) was built through min-max normalization and equal weighting. Mexico ranks fourth with 26.57 points, behind the Netherlands (92.14), New Zealand (60.91), and Argentina (55.72), and ahead of Kenya (15.42). Its main strength lies in access and affordability (81.2 % of the population using the Internet and an entry-level broadband basket costing 1.95 % of GNI per capita), while its main gap is in compute capacity, R&D investment, patents, and exports of digitally deliverable services. The analysis is complemented with correlations, K-means clustering, a weight-sensitivity test, and a reading through the 4C framework for artificial intelligence readiness. The report also documents the traceability and comparability limitations of the dataset, in particular the small sample ($n = 5$) and the non-automated capture of three of the four sources.

Index Terms

Digital economy, Digital Readiness Score, digital divide, AgTech, 4C framework, composite indicators.



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I. INTRODUCTION

The digital economy is no longer limited to manufacturing equipment or providing telephony: it includes a country's ability to use, produce, and export digital technology and services. In traditional sectors such as agriculture, that ability determines whether producers adopt precision-agriculture tools (sensors, telemetry, satellite imagery, data analytics) or depend on solutions developed abroad.

This report answers the central question of Activity 2 (Part I): *how prepared is each economy to participate in, generate, and capture value within the digital economy, and what are Mexico's main strengths and gaps relative to the other four economies on the team?* Team 3 (Agriculture) compares Mexico with the Netherlands, Kenya, Argentina, and New Zealand.

The work covers data acquisition and integration, the selection of eight indicators, the construction of the DRS, three analytical visualizations, a complementary analysis, a study of Mexico's digital gap, a reading through the 4C framework for artificial intelligence, and a final diagnosis. The whole workflow (acquisition, processing, and interactive dashboard) is implemented in the project repository to make it reproducible.

II. DATA ACQUISITION AND INTEGRATION

A. Sources and Procedure

The data come from four official organizations: the World Bank's World Development Indicators API, ITU DataHub, UNCTADstat, and WIPO IP Statistics. The script `01_fetch_data.py` queries the World Bank API for five of the eight indicators (period 2019–2024) and stores the unmodified response in `data/raw/`. The script `02_process_data.py` selects the year 2023, integrates the four sources into a single dataset (five economies $\times$ eight indicators), normalizes the values, and computes the DRS. Original and processed files are kept in separate folders.

B. Traceability

Table I summarizes the provenance register (`source_log.csv`); the full data dictionary is in `data_dictionary.csv`.

C. Temporal Comparability

The year 2023 was used for all eight indicators because it is the most recent year for which all five economies have data in the World Bank indicators (for 2024, for example, R&D spending is missing for Argentina, the Netherlands, and New Zealand). Using a common year avoids comparing, say,

a post-pandemic figure with a pre-pandemic one, which could create an apparent lag or improvement that actually reflects a different context. ICT services illustrate the risk: their share of Argentina's service exports goes from 14.1 % in 2019 to 24.6 % in 2021 and 15.9 % in 2023, so comparing different years would change the conclusion.

III. INDICATOR SELECTION

Exactly eight indicators were used, distributed according to the required quota: two for infrastructure or connectivity, one for access or usage, one for quality or affordability, two for digital economic activity, and two for technological capacity or innovation. They come from four official sources (the minimum required is three). Table II shows the observed values.

A. Exploration and Preparation

- **Missing data.** For 2023 there are no missing values in any of the eight indicators. Other years do have gaps (for example, Kenya's R&D before 2022 or New Zealand's in 2020 and 2022); they were not filled in. The script contains fallback values for R&D, but they are never triggered because all five economies have 2023 data.
- **Zeros versus unavailable data.** The zeros in the normalized matrix (e.g., Mexico's ICT service exports) do not indicate missing data: they correspond to the minimum observed among the five economies.
- **Scales and outliers.** Secure servers range from 297 to over 194,000 per million: the Netherlands is an extreme value that pushes the other economies close to zero after linear normalization. A logarithmic version was tested (Section V-C).
- **Inconsistencies.** The indicators mix units (percentages, rates per 100 and per million inhabitants), so normalization was required before aggregating them.

IV. CONCEPTUAL FOUNDATIONS

A. ICT Sector, Digital Economy, and Digitalized Economy

The **ICT sector** groups the industries that manufacture or provide information and communication technology goods and services (equipment, telecommunications, software). The **digital economy** covers activities whose business model depends on digital networks and platforms (cloud services, online marketplaces, mobile financial services). The **digitalized economy** describes the transformation of traditional sectors, such as agriculture, through the intensive adoption of digital technology.

TABLE I
PROVENANCE REGISTER OF THE EIGHT INDICATORS (OBSERVATION YEAR: 2023)

Code	Category	Organization / platform	Retrieval	Transformation
IT_NET_BBND	Infrastructure	ITU / World Bank (WDI)	JSON API	Selection of year 2023
IT_NET_SECR	Infrastructure	Netcraft / World Bank (WDI)	JSON API	Selection of year 2023
IT_NET_USER	Access / usage	ITU / World Bank (WDI)	JSON API	Selection of year 2023
ITU_PRICE_BASKET	Affordability	ITU DataHub	Entered in script*	Inverted scale in the DRS
ICT_SERV_EXP	Digital activity	IMF / UNCTAD / World Bank	JSON API	Selection of year 2023
DIGIT_DELIV_EXP	Digital activity	UNCTADstat	Entered in script*	Direct use of the percentage
RD_EXP_GDP	Technological capacity	UNESCO UIS / World Bank (WDI)	JSON API	Selection of year 2023
PATENT_RES_PM	Technological capacity	WIPO IP Statistics	Entered in script*	Patents per million inhabitants

* Values recorded directly in the script; see Section XI.

TABLE II
CONSOLIDATED INDICATOR MATRIX (2023)

Indicator	MEX	NLD	KEN	ARG	NZL
Fixed broadband (per 100)	20.75	43.26	2.39	25.36	37.85
Secure servers (per 1M)	412.1	194 962.9	297.1	5 451.2	18 993.7
Internet users (%)	81.18	97.01	32.07	89.23	93.33
Broadband basket (% GNI)	1.95	0.82	10.40	2.90	0.98
ICT service exports (%)	2.92	9.62	10.67	15.87	6.95
Dig. deliverable serv. (%)	24.5	58.4	39.2	64.2	44.8
R&D (% of GDP)	0.27	2.27	0.80	0.60	1.55
Resident patents (per 1M)	8.8	118.5	4.5	9.2	63.8

The indicator set used here best describes a **digitalized economy**: it does not only measure equipment supply or the size of the ICT sector, but also compute infrastructure, access, affordability, innovation capacity, and exports of digital services, which are conditions for sectors such as agriculture to become data-driven. It should be noted that the indicators are national and do not directly measure digital adoption on farms.

B. Growth of the Digital Economy

Three indicators represent different dimensions: (i) **Internet users** measure social adoption and potential market size; (ii) **secure servers** approximate the capacity to host and protect cloud workloads; (iii) **exports of digitally deliverable services** measure the ability to turn knowledge into foreign income. A single indicator is not enough: Mexico has 81.2 % Internet users but only 412 servers per million inhabitants, so a country can be a large consumer of digital services without being a producer.

C. Indicator, Evidence, and Interpretation

Observed data: the Netherlands records 194,963 secure servers per million inhabitants, versus 412 for Mexico. *Interpretation:* the indicator approximates the density of hosting and security infrastructure; the result is consistent with the Netherlands' role as a European traffic-exchange and data-center hub. *Supported conclusion:* the Netherlands has a much larger computing infrastructure base. *Unsupported conclusion:* that Dutch farmers are more profitable than Mexican farmers; profitability depends on land, subsidies, prices, and exchange rates, which this figure does not capture. In addition, the metric counts servers located in the country, not their agricultural use.

D. Digital Trade

An **e-commerce transaction** is the purchase or sale of a good or service carried out by electronic means; it may involve physical logistics (for example, buying tractor parts on an online marketplace). A **digitally delivered service** is a service whose provision takes place entirely over digital networks (for example, a crop-yield forecast based on satellite imagery delivered through an API). An online order of fertilizer is e-commerce but not a digitally delivered service, because the good is shipped physically.

V. DIGITAL READINESS SCORE

A. Methodology

The DRS is a composite indicator:

$$\text{DRS} = \sum_{i=1}^8 w_i I_{i,\text{norm}}, \quad \sum_{i=1}^8 w_i = 1, \quad w_i = \frac{1}{8}. \quad (1)$$

Normalization. Min–max scaling to $[0, 100]$ was applied using the minimum and maximum of the five economies:

$$I_{i,\text{norm}} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \times 100. \quad (2)$$

Direction of the variables. For the broadband basket (a higher cost means lower affordability) the scale is inverted: $(x_{\max} - x_i) / (x_{\max} - x_{\min}) \times 100$. **Weights.** Equal weights (0.125) were used because there is no empirical justification for favoring one dimension; this decision is tested in Section V-C. **Missing data.** There were no missing values in 2023, so no imputation was applied.

B. Results

C. Sensitivity Analysis

Because the DRS depends on the weights and the scale, the robustness of the ranking was checked with four weighting schemes (equal; emphasis on innovation, with $w = 0.25$ for R&D and patents; emphasis on exports, with $w = 0.25$ for the two digital-activity indicators; and emphasis on infrastructure, with $w = 0.25$ for broadband and servers), each with servers on a linear and on a logarithmic scale. The Netherlands ranks first in every case and New Zealand second, except with the export emphasis (Argentina second). Mexico stays fourth under the equal, innovation, and infrastructure schemes, and drops

TABLE III
NORMALIZED SCORES (0–100) AND DRS

Economy	Broadband	Servers	Users	Affordab.	ICT exp.	Dig. serv.	R&D	Patents	DRS	Rank
Netherlands	100.00	100.00	100.00	100.00	51.74	85.39	100.00	100.00	92.14	1
New Zealand	86.76	9.60	94.33	98.33	31.12	51.13	64.00	52.02	60.91	2
Argentina	56.20	2.65	88.02	78.29	100.00	100.00	16.50	4.12	55.72	3
Mexico	44.92	0.06	75.62	88.20	0.00	0.00	0.00	3.77	26.57	4
Kenya	0.00	0.00	0.00	0.00	59.85	37.03	26.50	0.00	15.42	5

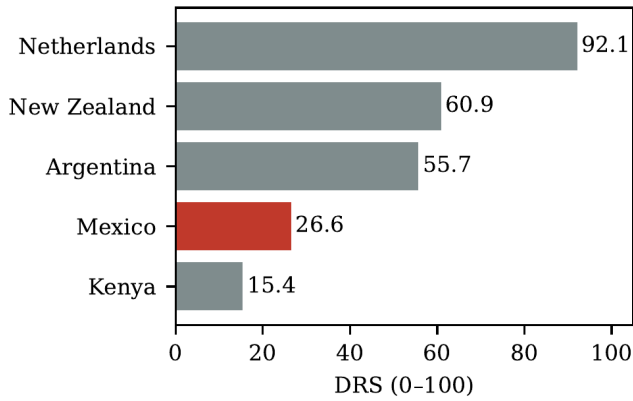


Fig. 1. Relative position of the five economies according to the DRS.

to fifth with the export emphasis (17.7 points, below Kenya's 26.4), which shows that its intermediate position depends on its performance in access and affordability. This qualifies the conclusion: fourth place is robust to the server scale, but not to every weighting.

VI. DATA VISUALIZATION

Three visualizations are presented, each answering a different question; the full interactive version (including a weight simulator) is in the dashboard `dashboard/index.html`.

A. Where does each economy stand?

What do I observe? Figure 1 shows a wide gap between the Netherlands (92.14) and the rest, and a separation of about 29 points between Mexico and Argentina. **What does it mean?** Mexico is closer to Kenya than to Argentina in the index. **What can I not conclude?** That Mexico is “not very digital” in absolute terms: the index is relative to this group of five.

B. What are Mexico's strengths and gaps?

What do I observe? In Figure 2, Mexico is below the average of its peers on most indicators; only Internet users and affordability are above it, and the largest shortfalls are in R&D, digitally deliverable services, and ICT exports. **What does it mean?** The gap is concentrated on the supply side (creating and exporting technology) rather than on demand. **What can I not conclude?** That Mexican agriculture is unproductive; the gap describes technology creation, not its use.

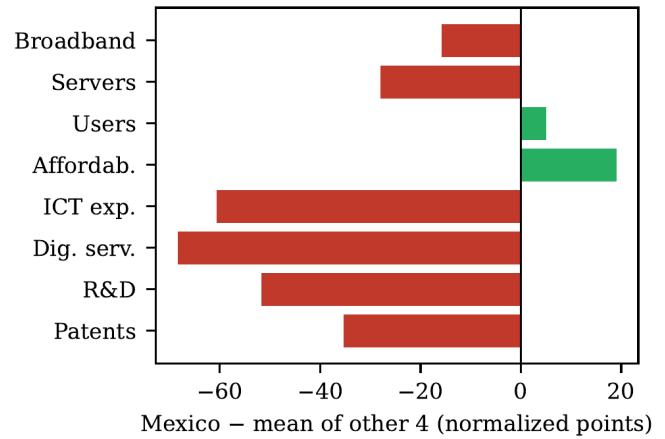


Fig. 2. Mexico's difference from the mean of the other four economies, in normalized points.

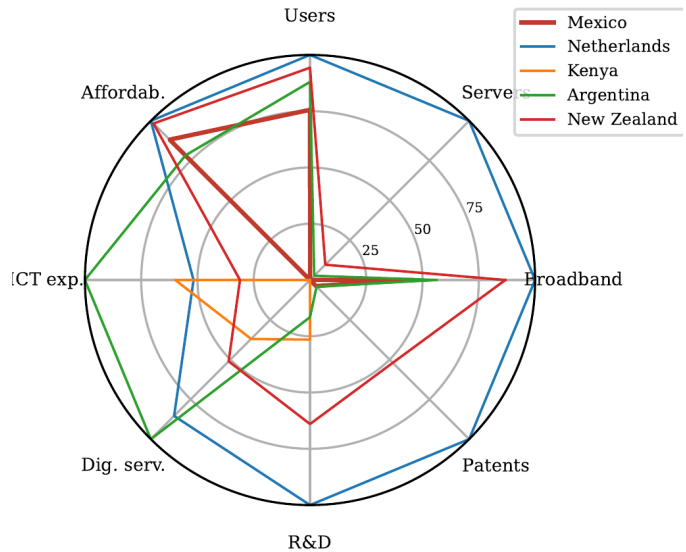


Fig. 3. Profile of the eight normalized indicators by economy (Mexico in red).

C. What profile differences does the DRS hide?

What do I observe? In Figure 3, Argentina and Kenya score high on service exports despite uneven infrastructure, while Mexico stands out in users and affordability. **What does it mean?** Two economies with similar DRS values can have opposite profiles: Argentina is a knowledge exporter; Mexico,

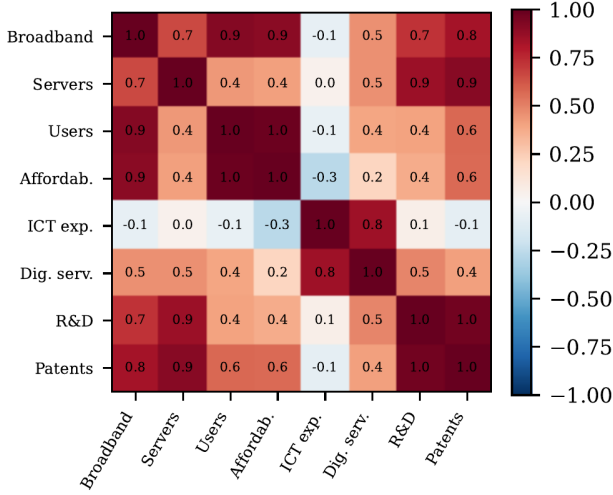


Fig. 4. Correlations between the normalized indicators.

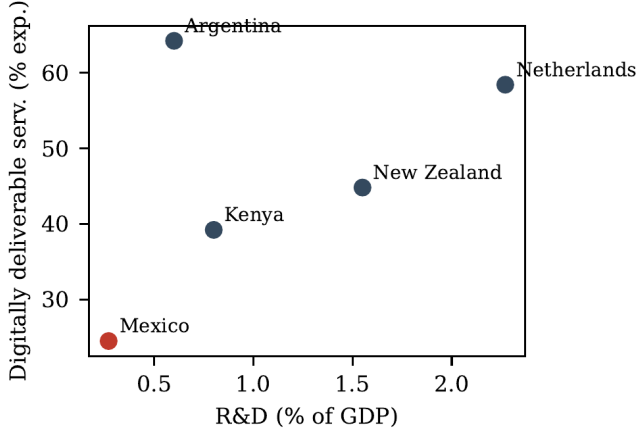


Fig. 5. R&D versus exports of digitally deliverable services.

a consumer. **What can I not conclude?** Which profile is preferable; it depends on national strategy.

VII. COMPLEMENTARY ANALYSIS

A. Relationships Between Variables

Pearson's correlation matrix was computed on the eight normalized indicators (Figure 4). The strongest relationships are between secure servers and patents ($r = 0.91$), between R&D and patents ($r = 0.96$), and between Internet users and the basket cost ($r = -0.98$ with the original cost: the lower the cost, the higher the use). In contrast, R&D and digitally deliverable services show a weak relationship ($r = 0.49$, $p = 0.41$; Figure 5).

Caution. With $n = 5$, correlations are unstable and a single country (the Netherlands) dominates several of them. Moreover, correlation does not imply causation: to argue that broadband cost causes greater use, one would need panel data over time, more countries, controls for income and education,

TABLE IV
RELATIONSHIP BETWEEN THE INDICATORS AND THE 4Cs

Dimension	Available indicators	Reading for Mexico
Connectivity	Broadband, users, affordability	Favorable access and price; intermediate fixed-broadband coverage
Compute	Secure servers	412 per million: very low versus the Netherlands (194,963)
Context	(none directly)	No data on open datasets, languages, or local applications
Competency	R&D, patents, service exports	0.27 % of GDP in R&D and 8.8 patents per million

and exogenous variation in prices (for example, regulatory changes).

B. Clustering

K-means with $k = 3$ was applied to the eight normalized indicators. The result places the Netherlands and Kenya in their own groups and groups Mexico, Argentina, and New Zealand together. With only five observations, the algorithm essentially isolates the extremes; clustering does not provide a robust typology, but it does confirm that Mexico, despite sharing a group with Argentina and New Zealand, differs by its consumption-oriented profile (Figure 3). The profile-based reading of the ranking therefore relies more on the radar analysis than on the clusters.

VIII. MEXICO'S DIGITAL GAP

Mexico's main gap is the **ability to generate and capture value**, not access to the network. By dimension:

- **Availability:** 20.75 fixed-broadband subscriptions per 100 inhabitants, versus 43.26 in the Netherlands.
- **Access:** 81.2 % of the population uses the Internet, below Argentina (89.2 %) and the high-income economies.
- **Affordability:** 1.95 % of GNI per capita, the most competitive among the middle-income economies in the group.
- **Quality:** the dataset includes neither speed nor latency; this is a limitation.
- **Capabilities:** 0.27 % of GDP in R&D and 8.8 resident patents per million, the lowest value in the group for R&D and one of the lowest for patents.
- **Outcomes:** 24.5 % of service exports are digitally deliverable and 2.9 % are ICT services, the lowest in the group.

High network coverage does not imply high utilization: coverage measures that infrastructure is available, whereas utilization depends on skills, content, devices, trust, and the existence of applications useful for the productive context. A producer may have a connection and use it only for messaging or entertainment. This is supported by the pattern in the data (high access, low production), but productive use could not be measured directly, which is another limitation of the dataset.

IX. AI READINESS: THE 4C FRAMEWORK

Is the information sufficient to claim that any country is prepared to develop and use advanced AI systems? No. The indicators only approximate Connectivity, Compute, and Competency indirectly, and there is none for Context. Secure servers with a certificate are not equivalent to compute capacity for training models, and patents and R&D are not AI-specific. Additional information would be needed on computing capacity (data centers, GPUs, cloud), availability of open agricultural data and of models in the local language and context, data-science and AI talent, actual technology adoption on farms, and data-governance frameworks. For this reason the framework is used here as a qualitative reading and not as a score.

X. FINAL DIAGNOSIS

- 1) **Position.** Mexico ranks 4th of five economies with a DRS of 26.57, below the Netherlands (92.14), New Zealand (60.91), and Argentina (55.72), and above only Kenya (15.42).
- 2) **Strength.** Access and affordability: 81.2 % Internet users and a broadband basket of 1.95 % of GNI per capita.
- 3) **Gap.** Capacity to produce and capture value: 412 secure servers per million, 0.27 % of GDP in R&D, 8.8 resident patents per million, and 24.5 % digitally deliverable services in service exports. Mexico behaves more as a consumer than as a creator of technology.
- 4) **Most interesting comparison.** Argentina: a Latin American upper-middle-income economy with 64.2 % digitally deliverable services and 15.9 % ICT services in its service exports, which suggests that the gap is not explained by income alone.
- 5) **Limitation.** The data are single-year national aggregates: they hide internal differences and do not measure agricultural adoption or AI compute. In addition, the DRS depends on normalization across only five countries and on the weights.

XI. LIMITATIONS

- **Traceability of three sources.** The ITU DataHub, UNCTAD-stat, and WIPO values are written directly in the acquisition script, and the `data/raw/` files for those sources are generated from them rather than downloaded unmodified from the official platforms. This conflicts with the provenance rule of the activity and must be corrected before delivery: download the official files, store them unchanged, and check the values against them.
- **Sample size.** With $n = 5$, correlations, clustering, and min-max normalization are sensitive to extreme values.
- **Single year.** Temporal evolution is not analyzed.
- **National aggregation.** The difference between technified agribusiness and rural smallholders is lost.
- **No measures of quality, productive use, or Context.**

XII. CONCLUSION

Mexico sits in the lower part of the group (fourth of five) because, although it has a solid base of access and affordable prices, its compute capacity, innovation, and exports of digitally

deliverable services are very limited. The result is robust to the server scale and to several weighting schemes, although Mexico drops to fifth place if digital exports are emphasized. The comparison with Argentina suggests that the gap can be narrowed through knowledge services without waiting to match the income of high-income countries, provided it is accompanied by investment in talent and R&D.

From a Data Engineering perspective, the project shows that a reliable diagnosis requires full traceability from the original data: the main pending improvement is to replace the manual capture of three sources with verifiable official downloads.

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